

Exploratory Data Analysis (EDA) on Titanic Dataset

1. Abstract

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process that helps in understanding the dataset, identifying trends, detecting anomalies, and discovering relationships among variables. In this project, the Titanic dataset is analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. The primary objective is to identify factors that influenced passenger survival and extract meaningful insights through statistical and visual exploration.

2. Introduction

The Titanic dataset is one of the most popular datasets used for data analysis and machine learning. It contains information about passengers aboard the RMS Titanic, including demographic details, ticket information, travel class, fare paid, and survival status.

The goal of this analysis is to:

- Understand the structure of the dataset.
- Identify missing values and data quality issues.
- Analyze survival patterns.
- Discover relationships among variables.
- Present findings through visualizations.

3. Dataset Description

The dataset contains passenger-related information such as:

Feature	Description
PassengerId	Unique passenger identifier
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class
Name	Passenger name
Sex	Gender
Age	Passenger age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Ticket	Ticket number
Fare	Ticket fare
Cabin	Cabin number
Embarked	Port of embarkation

Dataset Statistics

The dataset consists of passenger records with both numerical and categorical variables. Several columns contain missing values that require attention during analysis.

4. Data Cleaning and Preprocessing

Missing Values Analysis

The following columns contain missing values:

- Age
- Cabin
- Embarked

Observations

- Cabin has a significant number of missing values.
- Age contains several missing entries.
- Embarked contains only a few missing values.

These missing values can impact analysis and should be considered while interpreting results.

5. Exploratory Data Analysis

5.1 Survival Distribution

Observation:

The number of passengers who did not survive exceeded the number of survivors, indicating that survival was relatively rare during the disaster.

5.2 Gender Analysis

Observation:

Female passengers had a much higher survival rate than male passengers.

Interpretation:

This suggests that evacuation policies may have prioritized women during rescue operations.

5.3 Passenger Class Analysis

Observation:

Passengers traveling in first class were more likely to survive than those in lower classes.

Interpretation:

Higher-class passengers may have had better access to lifeboats and emergency assistance.

5.4 Age Distribution

Observation:

Most passengers belonged to the 20–40 age group.

Interpretation:

The passenger population was largely composed of young and middle-aged adults.

5.5 Fare Analysis

Observation:

Most passengers paid lower fares, while a small number paid premium fares.

Interpretation:

The fare distribution is highly skewed, reflecting different passenger classes.

6. Correlation and Relationship Analysis

Heatmap Analysis

Observation:

Fare and passenger class show a noticeable relationship with survival.

Interpretation:

Passengers who paid higher fares generally had better survival chances.

Pairplot Analysis

Observation:

Age shows weaker relationships with survival compared to fare and passenger class.

Interpretation:

Economic and travel-related factors appear to have had a stronger influence on survival than age.

7. Key Findings

1. Female passengers had significantly higher survival rates than male passengers.
2. First-class passengers were more likely to survive than second- and third-class passengers.
3. Most passengers were between 20 and 40 years old.
4. Fare and passenger class were strongly associated with survival probability.
5. The dataset contains missing values, particularly in the Cabin column.
6. Age showed a weaker impact on survival compared to gender and passenger class.
7. Higher ticket fares were generally associated with better survival chances.

8. Conclusion

The Exploratory Data Analysis of the Titanic dataset revealed several important factors influencing passenger survival. Gender emerged as one of the strongest predictors, with female passengers experiencing significantly higher survival rates. Passenger class and fare also played major roles, indicating that socioeconomic status influenced survival outcomes. While age contributed some information, its effect was less significant compared to gender and passenger class.

Overall, the project demonstrates how EDA techniques can be used to understand datasets, identify patterns, and generate meaningful insights that support further analysis and predictive modeling.